

# The Basic Trigonometric Identities (Homework)

Simplify.

①  $8\tan\theta + \csc\theta \tan\theta \sin\theta$

$9\tan\theta$

②  $\tan\beta + \frac{\sin\beta}{\cos\beta}$

$2\tan\beta$

③  $\frac{\cot w}{\cos w}$

$\csc w$

④  $(\sin\beta + 1)(1 - \sin\beta)$

$\cos^2\beta$

⑤  $(\sec A - 1)(1 + \sec A)$

$\tan^2 A$

⑥  $1 + \sec B \cot B \sin B$

$2$

⑦  $\frac{\csc\theta}{\cot\theta}$

$\sec\theta$

⑧  $\frac{\tan\alpha}{\cot\alpha}$

$\tan^2\alpha$

⑨  $\frac{\sin^2 w + \cos^2 w}{\tan^2 w + 1}$

$\cos^2 w$

⑩  $\sec^2 A (\tan^2 A + 1)$

$\sec^4 A$

⑪  $\sin^2\theta + \tan\theta \cot^3\theta + \cos^2\theta$

$\csc^2\theta$

⑫  $1 - \frac{\sin^2 w}{\tan^2 w}$

$\sin^2 w$

⑬  $\frac{\cos^2\alpha}{\sin^2\alpha} + 1$

$\csc^2\alpha$

⑭  $\frac{\sin^2\theta + \cos^2\theta + \cos\theta - 1}{\cot^2\theta + \sin\theta + 1 - \csc^2\theta}$

$\cot\theta$

⑮  $\frac{\tan^2\beta + 1}{\sec^2\beta}$

$1$

⑯  $\cot^2 w + \cos w \sec^2 w + 1 - \csc^2 w$

$\sec w$

⑰  $\frac{\csc^2 B}{\cot^2 B + 1}$

$1$

⑱  $\tan\theta \sin\theta \sec\theta$

$\tan^2\theta$

⑲  $\sin^2 A (\tan^2 A + 1)$

$\tan^2 A$

⑳  $(1 + \cot^2 B) \cos^2 B$

$\cot^2 B$